

Program : Diploma in Architecture	
Course Code : 4188	Course Title: Surveying Lab for Architecture
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To identify the accessories and perform chain survey, compass survey and plane table
- To conduct levelling practice and perform contouring survey
- To perform theodolite survey in the field and find the difference in elevations and distance between objects
- To utilize Total station survey in the field

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics -II	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Make use of survey instruments like chain, tape, ranging rod, cross staff and peg for calculating an area of small plot.	9	Applying
CO2	Apply appropriate equipment for levelling practice and performing contouring survey	9	Applying
CO3	Apply appropriate equipment for theodolite and perform theodolite survey in the field	9	Applying
CO4	Apply appropriate equipment for total station and perform total station survey in the field	12	Applying
	Series Test	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	2		2
CO2	3			3	2		2
CO3	3			3	2		2
CO4	3			3	2		2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Make use of survey instruments like chain, tape, ranging rod, cross staff and peg for calculating an area of small plot.		
M1.01	Determine the area of the plot by chain and cross-staff	3	Applying
M1.02	Determine the area of plot by compass surveying	3	Applying
M1.03	Define plane table survey	3	Understanding
CO2	Apply appropriate equipment for levelling practice and performing contouring survey		
M2.01	Determination of reduced levels by height of instrument method	3	Applying
M2.02	Determination of reduced levels by rise and fall method	3	Applying
M2.03	Determine levels and prepare contour plan	3	Understanding
	Series Test -I	3	
CO3	Apply appropriate equipment for theodolite and perform theodolite survey in the field		
M3.01	Determination of horizontal angle by repetition method	3	Applying
M3.02	Determination of horizontal angle by reiteration method	3	Applying
M3.03	Determination of horizontal distance and level difference between two points	3	Applying
CO4	Apply appropriate equipment for total station and perform total station survey in the field		
M4.01	Determination of horizontal angle by total station	3	Applying
M4.02	Determination of height of building by total station	3	Applying
M4.03	Conduct traverse using total station	6	Applying
	Series Test- II	3	

Text / Reference:

T/R	Book Title/Author
T1	Punmia, B.C, Jain, Ashok Kumar; Jain, Arun Kumar, Surveying I, Laxmi Publications, New Delhi.
R2	Kanetkar, T. P.; Kulkarni, S. V., Surveying and Levelling volume I, Pune Vidyarthi Gruh Prakashan.
R3	4. Duggal, S. K., Survey I, McGraw Hill Education, New Delhi.
R4	Saikia, M D.; Das. B.M.; Das. M.M., Surveying, PHI Learning, New Delhi.
R5	Subramanian, R., Fundamentals of Surveying and Levelling, Oxford University Press. New Delhi.
R6	Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning New Delhi.
R7	8. Bhavikatti, S. S., Surveying and Levelling, Volume 1, I. K. International, New Delhi. 9. Arora K R , Surveying Vol. I, Standard Book House

Online Resources:

Sl. No	Website Link
1	https://lecturenotes.in/subject/156/surveying-1-s-1
2	https://civiltoday.com/surveying/87-surveying-lecture-notes-pdf www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587 www.vssut.ac.in › lecture_notes › lecture1428642587
3	http://www.a-zshiksha.com/forum/viewtopic.php?f=149&t=61476
4	https://www.smartworld.com/notes/surveying-pdf-notes-lecture-notes-free-download/ www.alphace.ac.in › downloads › notes www.alphace.ac.in › downloads › notes www.alphace.ac.in › downloads › notes
5	https://edurev.in/studytube/Surveying--Part-1--Introduction-Notes--Surveying--/55e2c092-924a-4d87-95d4-62b01378460e_t?utm_campaign=&utm_source=&utm_medium=